Balaji Ramachandran

Software Engineer - IoT

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Professional Summary

A highly experienced software engineer with 8 years in product development for web, mobile applications, real-time IoT and AI solutions. Currently working as a researcher with multiple publications, I focus on developing edge-based IoT software solutions. I manage end-to-end projects and collaborate with diverse teams to design innovative IoT software solutions. I hold a Bachelor's in Electrical and Electronics and a Master's in Engineering Management. I am now seeking opportunities to further advance my expertise in software development, IoT and AI-driven solutions, particularly in a role that emphasizes technical leadership and business development.

Skills

Programming: Python, Java, PHP, .NET
Web and Mobile : HTML5, CSS, JavaScript, Android
Data Analysis: Pandas, NumPy, OpenCV, Scikit-learn, Matplotlib, Seaborn, Streamlit, Plotly, D3.js
Frameworks and Libraries: Flask, FastAPI,TensorFlow, Keras, PyTorch, TFLite, Selenium, Postman.
IoT Platforms: Thingsboard, Cumulocity,Node-Red, Azure IoT, Milesight IoT, Tuya.
Hardware Platforms: Raspberry Pi, Arduino, IIOT gateways,Linux, Jetson Nano
Cloud and DevOps: AWS, Azure, GCP, Docker, GitHub, Grafana, Jenkins
Machine Learning: AI frameworks, edge-based AI solutions, chatbots (OpenAI)
R&D: Research methodology, Technical writing, Prototype development and testing

Education

Middlesex University - United Arab Emirates — **Sep 2018 - Dec 2019**
Master of Science: Engineering Management — **Result:** First Class with Distinction

Anna University - India — **Aug 2012 - Apr 2016**
B.E: Electrical and Electronics Engineering, — **Result:** First class with Distinction

Experience

Research Engineer, — **Jul 2022- Present**
Dubai Electricity and Water Authority(DEWA) — Dubai, UAE

- Created over 15+ use cases for the utility industry using edge-based and AI solutions.
- Published 7+ papers in IEEE conferences and developed prototypes to advance IoT software.
- Led end-to-end project management, including feasibility studies and vendor collaboration.
- Trained 10+ interns and presented at events to share IoT expertise.
- Managed the development of an in-house IoT terminal, recognized as the Best Technical Project. The product is currently deployed, costs less than commercial solutions, and has the potential to save millions of dirhams if scaled.
- Facilitated communication between business users, vendors, and the product team, handling proposals and risk assessments.

Software Developer — **Jan 2019- Dec 2021**
Future Trend LLC — Dubai, UAE

- Executed projects related to Employee Attendance deployed and used by multiple government ministries of UAE.
- Worked as a Scrum Master to deliver Enterprise Projects as part of an Agile Team.
- Developed Backend applications to integrate SDKs of modern facial recognition and biometric devices.
- Provide Technical Leadership to the development Team.

Senior System Engineer **Jun 2016 - Aug 2018**
Infosys Limited Trivandrum, India

- Implemented development, validation, and support activities in line with architecture requirements for various software tools related to automation and web development.
- Designed, developed, and deployed backend services with high availability and scalability.
- Developed web applications that handle high volumes of data and backend services.
- Applied both test-driven and behavior-driven development by following the Software Testing Lifecycle.
- Emphasized cloud deployment and DevOps practices to enhance the efficiency and reliability of software delivery processes.

PROJECTS

Smart Crew Dispatch: Developed an end-to-end dashboard that integrates real-time sensor readings and geolocation from sensors installed in 10 field vehicles, resulting in a 20% increase in operational efficiency for crew dispatch.

Energy Disaggregation: Designed and managed the end-to-end development of a custom edge device for smart energy monitoring, installed in 30 homes alongside COTS products.The edge devices, operational for the last two years, feature AI algorithms that provide high-frequency disaggregation of appliance energy usage with an accuracy of 80%, supported by web and mobile applications for seamless monitoring.

Power Quality Monitoring: Developed an IoT-based system for real-time monitoring and analysis of power quality parameters, including power harmonic filters and power quality analyzers, through LTE CAT-M communications. Deployed 10 devices with edge capabilities, eliminating offline monitoring and enabling the system to raise automated alarms during abnormal events, significantly increasing operational efficiency.

Water Meter Monitoring: Designed an IoT solution for remote water meter monitoring, enabling accurate billing and efficient resource management.Developed an NB-IoT and LoRaWAN-based solution with an IP67 rated enclosure, managing data integration and device management functionality for enhanced performance and reliability.

Smart IoT Device with Mobile App: Developed a mobile app and smart device for field crews, enabling real-time data access and task management.Integrated thermal cameras, Bluetooth-enabled measurement devices, and AI algorithms for object and anomaly detection, along with location tracking. Condition monitoring speed improved from 20 minutes to 5 minutes per substation, enhancing efficiency.

Decision Support System: Developed software architecture for load forecasting and historical data analysis using 10,000 customer energy meter data from the last 5 years.Integrated AI algorithms for energy forecasting and a dashboard displaying energy KPIs, allowing users to compare performance at the region, community, and building levels. The tool is optimized to handle big data efficiently.

IoT Terminal: Managed the end-to-end product development lifecycle of an in-house IoT terminal, ensuring compliance and integration with various sensor interfaces.

The terminal supports multiple communication technologies (LoRa satellite, LoRaWAN, LTE CAT-M, NB-IoT, Wi-Fi, Bluetooth) and different sensor interfaces (RS485, GPIO). It can be solar-powered, with custom firmware tailored for industrial IoT solutions.

Green Hydrogen Monitoring: Created a remote dashboard for battery systems at MENA's first Green Hydrogen plant, using Tesla lithium-ion (1.21 MW) and NaS (1.2 MW) technologies to display battery state of charge (SOC) for hydrogen production control.

Load Forecasting Dashboard: Developed a Python-based web dashboard that integrates AI models to forecast feeder loads for distribution planning using historical data from 20 substations collected over 5 years. The interactive dashboard employs advanced ML techniques, featuring maps and user-friendly displays of ML metrics.

3U Nano-Satellite Integration: Implemented a centralized system for IoT terminals using LoRa communication, enabling connectivity with the utility's first nano-satellite, DEWASAT-1, and successfully demonstrated three utility use cases; the project is now scaling up with other satellite providers.

Substation Monitoring: Substation Monitoring: Developed an IoT platform for telemetry from electrical monitoring devices for feeders and environmental monitoring devices, currently deployed in 25 substations with high data transmission rates, and working on scaling to 1,000+ substations.

ERP for SMEs: ERP for SMEs: Developed a multi-platform ERP app utilized by 10+ SMEs, featuring geo-fencing and payment integration, resulting in a 30% increase in operational efficiency.

Attendance App for NOC UAE: Built a geo-fenced attendance app with multi-language support, successfully tracking attendance for 500+ employees across 3 locations.

Face Recognition Integration: Developed a Python tool for a biometric-based attendance and security system for a semi-government media agency, enabling fast individual recognition for 200+ employees.

IPTV Automation: Created over 100 automation test scripts for a telecom operator's digital TV services, automating the STB testing process using a TDD model with an STB tester.

Revenue Analytics for Game Developer: Developed a web app that analyzes millions in revenue from multiple ad networks, providing insights for each game application and optimizing revenue strategies.

Publications

Ramachandran, B. et al., IoT Enabled Power Quality Monitoring in Substations via LoRa LEO Satellite Communication, EECCIS 2024.

Chikte, R., Ramachandran, B., et al., Short-Term Energy Forecasting for Smart Buildings Using ML Models, ASME IMECE 2024.

Ramachandran, B., Bendoukha, S. A., & Karunamurthy, J. V., Direct Satellite Communication Improvement Using LoRa-FHSS (DEWASAT-1), MTTW 2023.

Ramachandran, B., et al., IoT-Based Test Facility for Water Transmission Line with Leak Simulation, ITIKD 2023.

Haitaamar, Z. N., Ramachandran, B., & Karunamurthy, J. V., IoT-Enabled Laboratory Design & Implementation, ITIKD 2023.

Subramanian, V., Ramachandran, B., & Karunamurthy, J. V., Hardware Doppler Shift Emulation for LoRa LEO Satellite Communication, ITIKD 2023.

Certifications

Scrum Master — Oct 2021

Awards

Best Collaborator Award (2025): R&D Center – Cross-department Collaboration
Quality Achievement Award (2025): CE Certification – IoT Terminal
Nujoom Award (2024) : Best Technical Project – Omnihub IoT Terminal
Best Paper Award(2024): PQA Monitoring – EECCIS Conference
Best Ideation Session Award (2024): Innovation & Invention Disclosures

Personal Details

Date of Birth: 15/06/1995

Nationality: Indian

Visa Status: UAE Golden Visa

Languages: English and Tamil